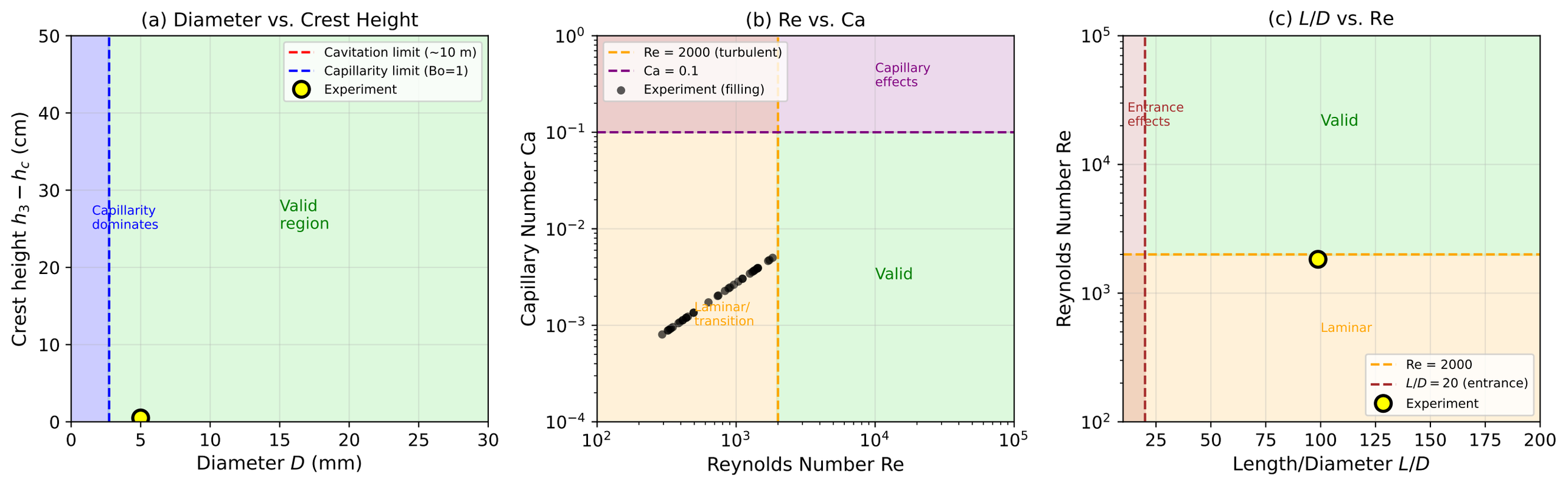
(a) Diameter vs. Crest Height
Cavitation limit (~10 m)
Capillarity limit (Bo=1)
Experiment
Capillarity dominates
Valid region
Crest height $h_3 - h_c$ (cm)
Diameter D (mm)
(b) Re vs. Ca
Re = 2000 (turbulent)
Ca = 0.1
Experiment (filling)
Capillary effects
Valid
Laminar/ transition
Capillary Number Ca
Reynolds Number Re
(c) L/D vs. Re
Entrance effects
Valid
Laminar
Re = 2000
L/D = 20 (entrance)
Experiment
Reynolds Number Re
Length/Diameter L/D